

1 / Shared lab stocking strategy

Purchase proposal | September 14, 2026 | For September 15 discussion. Scope: general embedded electronics and avionics prototyping across multiple projects, plus a separate PDB project kit. Quantities below are recommended initial stock targets, not an assumed board count or an approved purchase order.

Priority	What to stock	Starting policy
P1 - core stock	Frequently reused passives, indicator LEDs, small transistors, common logic and analog ICs, basic connectors.	Order after Ryan's inventory and machine/carrier check. Keep stocked through reorder thresholds.
P2 - expand on demand	CAN/RS-485, memory, MCUs, selected regulators, precision analog and uncommon passive values.	Small quantities once the lab chooses a preferred family; expand when recurring designs use it.
Project kits	PDB protection controller, power MOSFETs, shunt, fuse and other board-specific selections.	Reserve by BOM and build batch. Do not count committed project stock as freely available lab stock.
Equipment and process	Feeders, nozzles, dry/ESD storage, paste, stencil printing, reflow, inspection and harness tooling.	Confirm what comes with the machine and what the lab already owns before quoting gaps.

Package standard

Proposed default for NEW general-purpose designs: 0603 EIA passives. Keep an 0402 satellite stock for compact Osiris work once the machine is confirmed capable. Keep 0805/1206 where power, capacitance or voltage requires them. This policy does not change any existing PCB footprint.

Ryan's books

Inventory the books by value, package, rating, quantity, MPN and carrier. Keep assorted values for tuning and hand rework. Count them as machine stock only if identifiable and feedable. A book does not by itself replace continuous tape/reels.

What makes this save time

Use a shared approved-parts library with stock IDs, footprints, datasheets and known carrier settings. Design new boards around these parts when electrically appropriate; reuse validated circuit blocks. Avoid a different regulator, connector and package on every project.

Quantity interpretation

Targets are PER value, color, voltage option or selected IC variant unless stated otherwise. Subtract verified usable stock. Allow separate feeder setup/repair stock. Prices, supplier stock, factory reel quantities and packaging fees need quotes; no spend total is claimed.

2 / Passives: the reel inventory

P1 means the first stocking wave. Prefer one approved manufacturer part per value/specification/package. Full reels suit sustained demand; custom reels provide smaller machine-ready stock. No automatic substitution into the PDB or Osiris is implied.

Priority / group	Values and minimum selection criteria	Target per variant
P1 / resistors, 0603	0 ohm jumpers; 100 ohm, 330 ohm, 1 k, 4.7 k, 10 k, 100 k. 1% for nonzero values; at least 0.1 W nominal with application derating.	1,000 each; quote one factory reel as alternative
P1 / resistors, 0603	10, 22, 33, 47, 49.9, 220, 470 ohm; 2.2 k, 3.3 k, 6.8 k, 22 k, 47 k, 1 M. Same 1% class.	250 each
P1 / ceramic caps, 0603	100 nF, X7R, >=25 V, 10%; 1 uF, X7R, >=16 V, 10%. Choose exact MPN after bias/rating review.	2,000 of 100 nF; 1,000 of 1 uF
P1 / ceramic caps, 0603	1 nF and 10 nF, X7R, >=50 V, 10%.	250 each
P1 / bulk ceramic caps	4.7 uF and 10 uF, X7R, >=16 V, 0805. Verify effective capacitance at working voltage.	250 each
P2 / signal capacitors	22 pF, 47 pF, 100 pF, 1 nF, C0G, >=50 V, 0603; tolerance selected for the application. Crystal load values are circuit-specific.	100 each
P2 / compact 0402 stock	0 ohm, 1 k, 4.7 k, 10 k, 100 k; 100 nF X7R >=16 V. Exact ratings and MPNs reviewed separately.	500 each resistor; 1,000 caps; after machine check
P2 / larger power passives	PDB 0805/1206 parts use the exact frozen order list. General bulk caps, shunts and precision dividers selected per power circuit.	Project quantities; promote proven repeat parts later
P2 / ferrites and inductors	Start a small evaluation assortment; choose ferrite impedance curve/DC current/DCR and inductor inductance/saturation/RMS current/footprint together.	10-25 per selected MPN; no generic inductor reel buy

Avoid unnecessary breadth

Do not buy full reels of every E-series value. Ryan's books can cover experiments; machine-ready stock should favor repeat values. The targets above establish useful depth without claiming every item is already shared across project BOMs.

Reel quotation

Ask for target quantity on a custom reel AND factory reel total price, including reeling fees, minimum packs and available continuous length. Final manufacturer part numbers and carrier dimensions must appear on the order.

3 / Common ICs: a reusable shelf

These are function/family recommendations checked against manufacturer product pages. Family names are NOT complete order codes. Select one package and full ordering suffix per row before purchase; maintain it in the lab CAD library. Start with small custom reels or supported continuous cut tape.

Priority / function	Candidate family and selection boundary	Target
P1 / 3.3 V and 1.8 V LDO	TI TLV755P: select fixed output and SOT-23-5 variant. Low-voltage local rail use; 5.5 V maximum input, NOT a direct 4S battery regulator.	50 of 3.3 V; 25 of 1.8 V
P1 / dual op amp	TI TLV9002: general low-voltage signal conditioning. Prefer SOIC-8 for initial shared library if board area permits. Review input/output range and bandwidth per use.	25
P1 / comparator	TI TLV3201: threshold detection. Choose exact package, verify supply/input range and add designed hysteresis where required.	25
P1 / Schmitt inverter	TI SN74LVC1G14: signal cleanup / logic inversion; select SOT-23 variant and verify logic thresholds.	50
P1 / tri-state buffer	TI SN74LVC1G125: controlled signal buffering. Select SOT-23 variant; check enable and power-off behavior.	50
P1 / direction-controlled translator	TI SN74LVC1T45: push-pull digital level translation. Not a universal bidirectional I2C translator.	25
P1 / load switch	TI TPS22918: local low-voltage rail switching, up to 5.5 V input. This is not a 4S battery protector or a current-limiting eFuse.	25
P2 / current-sense amplifier	TI INA180: analog current measurement. Choose gain/package and common-mode range for the circuit. PDB INA228 remains a separate exact BOM part.	25 of one gain variant
P2 / CAN transceiver	TI TCAN332: candidate for 3.3 V CAN designs. Select exact variant/package against required bus mode and protection.	20 if CAN is a recurring lab interface
P2 / platform-dependent ICs	Choose ONE familiar MCU platform plus supported programmer; then matching flash/EEPROM, RS-485, I2C translator, reset supervisor and USB-UART parts as projects require.	10-20 per selected MPN; buy after platform review
P2 / switching power	Choose a small number of validated buck-regulator circuits with matching inductors, caps and layout. New Rev B power design is not yet selected.	10-20 matched circuit kits after design selection

Design once, reuse

For each stocked IC: store symbol, footprint, pin mapping, decoupling requirements and a tested example circuit. Order by complete manufacturer code, not just a familiar name; gain, voltage, pinout and packaging variants matter.

Upgrade to larger reels

When repeated builds establish demand, increase the selected IC stock target. Keep expensive/specialized devices at small quantities until recurring consumption justifies more.

4 / Discretes, connectors and tools

This list fills common gaps that delay a build even when its ICs are in stock. Quantities are proposed shared-stock targets. Exact device ratings, mechanical drawings and packaging must be selected before ordering.

Priority / category	What to request	Target
P1 / indicator LEDs	Red, green, blue; 0603, identified polarity and brightness/current specification.	250 per color
P1 / small transistors	One NPN and one PNP switching transistor in SOT-23; one small logic-level N-MOSFET and one P-MOSFET. Verify pinout and MOSFET RDS(on) at actual gate voltage.	100 per selected MPN
P1 / diodes	One fast small-signal diode and one small Schottky in chosen standard packages. Size power Schottky, TVS and ESD arrays separately for their circuits.	100 per small diode MPN
P2 / interface protection	One selected low-capacitance ESD array for the actual interface voltage and speed; TVS parts selected for the power bus.	25-50 per selected MPN
P1 / simple interconnect	2.54 mm breakaway male/female headers and jumpers; test points and chosen small tactile switch. Headers generally hand/through-hole assembly.	25 header strips per type; 100 jumpers; 100 test points; 50 switches
P1 / harness standard	Choose one locking wire-to-board family. JST GH is a candidate already used in PDB references. Start with 2/3/4/6-position board connectors, matching housings and contacts.	25 per chosen connector/housing variant; 500 contacts
P2 / project connectors	Power connectors, USB, SWD/programming connector and RF connectors must match mechanical plans. Quote board connector AND mating cable/housing.	10-25 per selected variant
P1 / feeder readiness	Compatible feeders, nozzles, cut-tape fixtures if supported, splice/leader supplies, reel storage and durable labels. Width/pitch and slots from selected machine.	Quote machine-specific kit; do not assume counts
P1 / assembly process	Paste compatible with process; stencil printing/support, reflow oven/profile measurement, microscope, solder/rework tools, tweezers and flux/cleaning supplies.	Inventory equipment; buy gaps and consumables
P1 / handling and harness work	ESD work area, component storage per moisture requirements, stock bins, wire, heat-shrink, matched crimper and extraction tools.	One shared setup; replenish consumables

Do not buy short-lived supplies in bulk

Buy solder paste to the near-term assembly schedule and manufacturer storage/shelf-life requirements. Stencils are specific to released PCB/panel layouts.

Machine-ready does not mean machine-only

Many headers and power connectors need hand or through-hole assembly. Reserve a separate work step and tools; verify actual machine capabilities for any nonstandard carrier.

5 / Purchasing and replenishment

The aim is to keep recurring builds moving. A shared inventory plus standard parts and repeatable feeder setups will save more time than an untracked assortment of reels.

Step / owner to assign	Deliverable	Completion gate
1 / Lab inventory owner	Count Ryan's books, reels, connectors and equipment; identify unknown parts and reserved project stock.	Usable free stock known per exact MPN/package/carrier.
2 / Machine owner	Document supported feeders, reel sizes, nozzles, smallest packages and supplied accessories.	Stock packaging and equipment quote match the machine.
3 / Hardware lead	Approve package defaults and full MPNs for P1 groups. Use the family shortlist on page 3; resolve exact variants.	Each order line has a complete code/spec, datasheet and footprint.
4 / Purchasing	Quote P1 first; quote P2 separately. Compare custom versus factory reels on TOTAL price and availability.	Budget-approved quantities net of stock and reservations.
5 / Receiving / assembly	Label MPN, value, package, lot, quantity, location, moisture status and carrier; verify identity before stocking.	Trial loading and a first-article board establish feeder/vision/reflow settings.
6 / All designers	Use approved stock parts in new designs where appropriate; reserve material at design release.	No unrecorded borrowing from project kits.

Starter reorder rule

Until usage data exists, review weekly and flag core passives at 25% of target or below; flag core ICs at 10 units or below. Reorder to target after subtracting open purchase orders. These are provisional rules, not measured consumption forecasts.

Once consumption is known

Reorder point = expected use during supplier lead time + safety stock. Available stock = on-hand usable stock minus project reservations. Include setup losses in consumption and increase safety stock for long or variable lead times.

Tomorrow's boss request

Approve a shared P1 stock budget separate from project BOM purchases, assign the inventory/machine owners, and authorize distributor quotes. P2 expands after lab platform choices. The existing 27-code PDB list remains a separate procurement attachment.

Sources / checked September 14, 2026

Manufacturer families: <https://www.ti.com/product/TLV755P> ; /TLV9002 ; /TLV3201 ; /SN74LVC1G14 ; /SN74LVC1G125 ; /SN74LVC1T45 ; /TPS22918 ; /INA180 ; /TCAN332 (all on www.ti.com/product/). Packaging: <https://www.digikey.com/en/resources/product-services> and <https://forum.digikey.com/t/details-about-digi-reels/1303> . Stock quantities and priorities are planning recommendations, not manufacturer guidance.